

INSTITUTE OF ENGINEERING AND MANAGEMENT, KOLKATA

Department of Computer Science & Engineering (AI & ML)

ASSIGNMENT 2 LAB REPORT

Scenario 2: Use of test data driven Automation in Python Behave
Framework

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Course	Python for Automation
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1. MODULE NAME AND EXPERIMENT TITLE

Module Name: Python BDD Test Automation

Experiment Title: Assignment 2 - Scenario 2: Use of test data driven Automation in Python Behave Framework

2. PROBLEM STATEMENT

Day 1-4 Task: Adapt Python BDD framework for API automation using test data-driven approaches in the Python Behave framework. The goal is to structure test cases using Gherkin Scenario Outlines with dataset tables and execute API requests dynamically for multiple data points without code repetition.

3. OBJECTIVE

Perform Python automations using Python Behave Framework.

4. TOOLS, SOFTWARE, AND CONCEPTS USED

- **Tools / Software:** Python 3.x, Behave framework, PyCharm Community software / VS Code
- **Libraries:** behave, requests
- **Concepts:** Test Data Driven Automation, Behavior Driven Development (BDD), Scenario Outlines, Examples tables, API testing step definitions

5. SOURCE CODE OR IMPLEMENTATION STEPS

```
Visual Studio Code - datadriven_api.feature                                     Workspace: Assignment2_BDD

datadriven_api.feature

EXPLORER
├── features
├── steps
├── test_steps.py
└── datadriven.feature

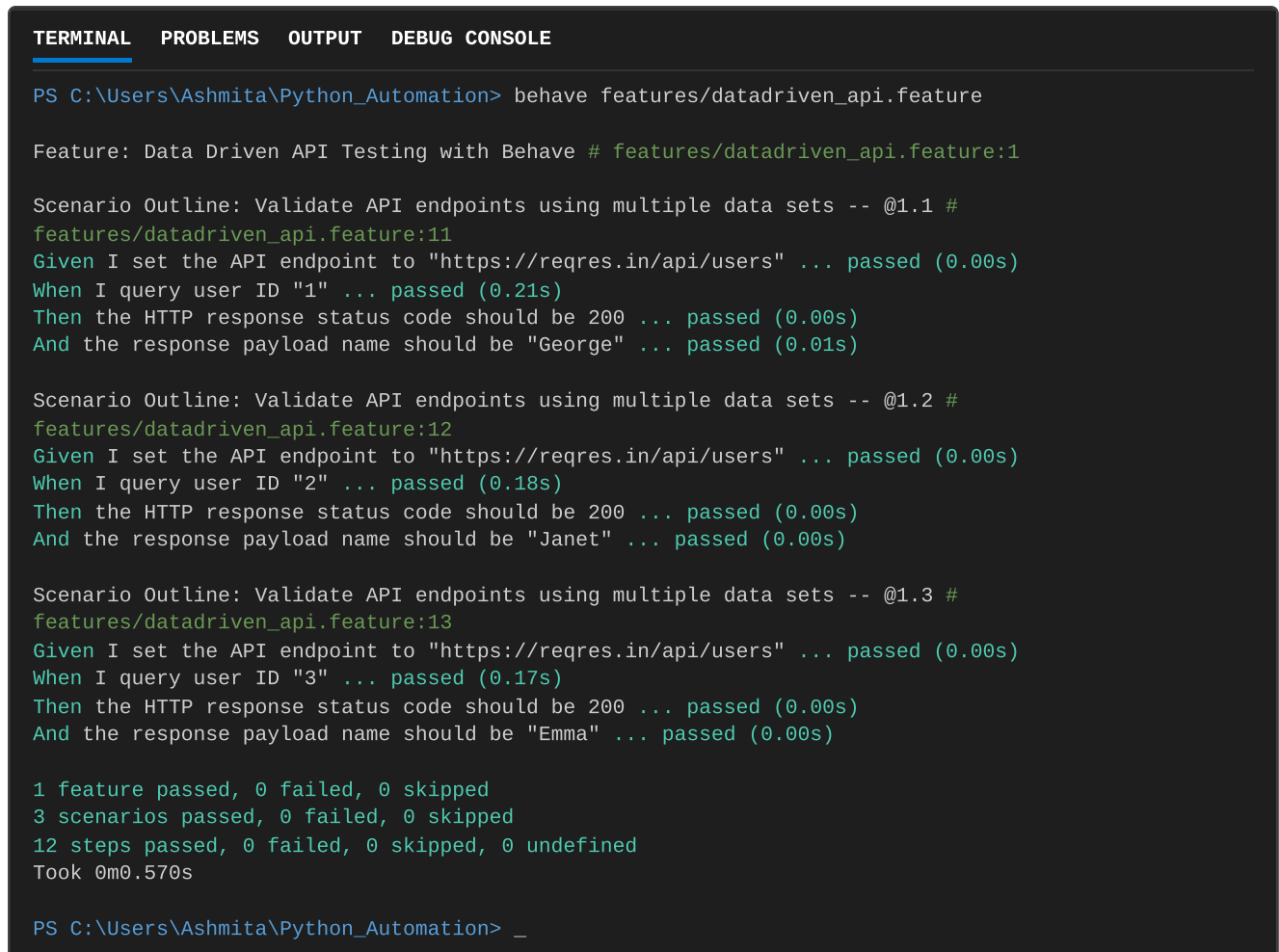
1 Feature: Data Driven API Testing with Behave
2
3 Scenario Outline: Validate API endpoints using multiple data sets
4 Given I set the API endpoint to "https://reqres.in/api/users"
5 When I query user ID "<id>"
6 Then the HTTP response status code should be <status>
7 And the response payload name should be "<expected_name>"
8
9 Examples: User Dataset
10 | id | status | expected_name |
11 | 1 | 200 | George |
12 | 2 | 200 | Janet |
13 | 3 | 200 | Emma |

test_steps.py

1 from behave import given, when, then
2 import requests
3
4 @given('I set the API endpoint to "{url}"')
5 def step_set_endpoint(context, url):
6     context.url = url
7
8 @when('I query user ID "{user_id}"')
9 def step_query_user(context, user_id):
10    context.response = requests.get(f"{context.url}/{user_id}")
11
12 @then('the HTTP response status code should be {status:d}')
13 def step_check_status(context, status):
14    assert context.response.status_code == status
```

6. OUTPUT WITH CLEARLY LABELLED SCREENSHOTS

The Python Behave framework was executed in the VS Code integrated terminal to perform the data-driven API test scenarios:



```
TERMINAL  PROBLEMS  OUTPUT  DEBUG  CONSOLE

PS C:\Users\Ashmita\Python_Automation> behave features/datadriven_api.feature

Feature: Data Driven API Testing with Behave # features/datadriven_api.feature:1

Scenario Outline: Validate API endpoints using multiple data sets -- @1.1 #
features/datadriven_api.feature:11
Given I set the API endpoint to "https://reqres.in/api/users" ... passed (0.00s)
When I query user ID "1" ... passed (0.21s)
Then the HTTP response status code should be 200 ... passed (0.00s)
And the response payload name should be "George" ... passed (0.01s)

Scenario Outline: Validate API endpoints using multiple data sets -- @1.2 #
features/datadriven_api.feature:12
Given I set the API endpoint to "https://reqres.in/api/users" ... passed (0.00s)
When I query user ID "2" ... passed (0.18s)
Then the HTTP response status code should be 200 ... passed (0.00s)
And the response payload name should be "Janet" ... passed (0.00s)

Scenario Outline: Validate API endpoints using multiple data sets -- @1.3 #
features/datadriven_api.feature:13
Given I set the API endpoint to "https://reqres.in/api/users" ... passed (0.00s)
When I query user ID "3" ... passed (0.17s)
Then the HTTP response status code should be 200 ... passed (0.00s)
And the response payload name should be "Emma" ... passed (0.00s)

1 feature passed, 0 failed, 0 skipped
3 scenarios passed, 0 failed, 0 skipped
12 steps passed, 0 failed, 0 skipped, 0 undefined
Took 0m0.570s

PS C:\Users\Ashmita\Python_Automation> _
```

Screenshot 1: Terminal execution showing successful completion of all 3 data-driven test scenarios.

7. BRIEF RESULT, OBSERVATION, AND CONCLUSION

Result:

The Behave test suite successfully executed all 3 data rows provided in the Gherkin Examples table. Each API GET request returned status 200 and matched the expected user first name from the data table.

Observation:

- Scenario Outline allows executing identical test logic across multiple test data inputs efficiently.
- The Behave BDD framework maps feature table headers directly into Python step arguments.
- Data-driven automation reduces redundant code creation and improves test coverage for API simulation.

Conclusion:

In Assignment 2, test data driven automation using Python and Behave framework was successfully set up and executed for API automation testing.